

Jade Clement

UofT Engineering Science Student

Ambitious and enthusiastic Machine Intelligence graduate in 2026 with research and production experience eager to make an impact on real world applications of engineering

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EDUCATION

Engineering Science: Machine Intelligence

University of Toronto

09/2020 - 05/2026

- MIE429: Machine Intelligence Capstone Design
- ECE324: Machine Intelligence, Software and Neural Networks
- CSC413: Neural Networks and Deep Learning
- ECE353: Systems Software
- ECE358: Foundations of Algorithms (Data Structures)
- MIE424: Optimization in Machine Learning
- ROB311: Artificial Intelligence
- ECE421: Introduction to Machine Learning
- MIE451: Decision Support Systems
- ECE368: Probabilistic Reasoning
- CSC343: Introduction to Databases
- ECE419: Distributed Systems

PROJECTS

Full Stack Web/Mobile Application (08-09/2025)

- Deployed a **distributed full-stack application** (React.js, Node.js/Express, PostgreSQL) serving **60+ active users**
- Available on the web and App Store (soon to Google Play)
- Integrated **AWS S3** and **Cloudfront** for scalable image storage and persistence data management.
- Built a **RAG Q&A** system using Pinecone and OpenAI to answer club-related questions from indexed website content
- Developed real-time workout signup and waitlist system with automated email notifications.
- Built an admin dashboard for member management, attendance tracking and analytics.

ML Document Classification (10/2023)

- Implemented classifiers with different encodings (binary, tf-idf) using **scikit-learn**, **numpy**, **pandas**, and **scipy** libraries
- Varied the feature set, feature encoding, and hyperparameters to optimize performance on classifying 20,000 newsgroup documents

Capstone Project (08/2023 - 12/2023)

- Collaborated with teammates and clients at Perimeter AI to brainstorm solutions to their data imbalance problem and ultimately help them improve their AI algorithm that detects cancer on OCT scans
- Explored techniques such as CGANs, Object Detection and Elastic Transform to generate new data leading to a **5% testing accuracy increase on ResNet-18**

Natural Language Processing Projects (10/2023 - 12/2023)

- Performed **sentiment analysis** on hotel review data scraped from Tripadvisor and plotted output to visualize results
- Designed a **prompt engineering** experiment to evaluate various prompt styles for a pretrained LLM on a task involving extracting specific information from email chains

Web Development (06/2023 - 08/2023)

- Programmed and designed my personal website using HTML, CSS and Javascript which can be found at jadeclement.com

Self-balancing Robot (01/2022 - 04/2022)

- Brainstormed, planned, designed and built a self-balancing robot prototype with 4 team members

SKILLS

Python ML/DL/RL MatLab C HTML CSS Git

JavaScript PyTorch Tensorflow Keras Node.js

Data Structures Communication PostgreSQL Django

EXPERIENCE

LLM Reasoning and Faithfulness Research

Human-Centered Data Science Lab

Toronto, ON

08/2025 - Present

- Designing an evaluation pipeline and web platform application for Large Language Models to assess reasoning quality on harmful text datasets.
- Implementing a modular back end to run multi-stage reasoning analysis and quantify coherence, sufficiency, necessity, and reliance in model explanations.
- Developing interactive visualizations and quantitative metrics to promote faithful reasoning and advance AI safety tools.

Software Engineering Intern

SingleKey

Toronto, ON

05/2024 - 04/2025

- Developed backend features using **Django** and **PostgreSQL**
- Built **full-stack** features using **Python** and **Node.js**
- Contributed to a robust and maintainable codebase through active participation in code reviews
- Researched emerging technologies to support future platform enhancements, including AI initiatives

Machine Learning Research Assistant

Robotics and Control Lab

Vancouver, BC

05/2023 - 08/2023

- Reviewed academic papers on generative models, especially **diffusion models**
- Designed and planned a machine learning model to generate echocardiogram videos by collaborating with team members
- Programmed and debugged in Python within Docker containers to reproduce results of the paper in the same field

Medical Imaging Research Assistant

UBC Centre for Heart & Lung Innovation

Vancouver, BC

05/2021 - 08/2021

- Analyzed **50+ cardiac CT cases** to investigate cardiovascular biomarkers derived from lung CT data.
- Resolved technical issues in 3D lung reconstruction software and supported quantitative image analysis workflows.
- Presented findings to a large audience of research professionals and clinicians, concluding that **quantitative lung analysis** is feasible within the cardiac CT field of view.

AWARDS AND LEADERSHIP

President of the University of Toronto Triathlon Club (2025-Present)

Qualifier for the Worlds Triathlon Championship (2024 & 2026)

Dean's Merit Award from the University of Toronto (09/2020)

Volunteer at Google Girl-Powered Robotics(2020)

LANGUAGES

English (native), French (fluent), Spanish (beginner)